

DRESDEN, Germany

WMO: 104880

Lat: **51.1281N**Lon: **13.7544E**Elev: **762**StdP: **14.30**Time Zone: **1.00 (EUC)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	17.2	5.3	7.4	14.0	10.4	9.6	19.6	25.1	39.6	22.9	37.9	7.2	70	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	87.7	66.7	84.0	65.2	80.5	64.7	68.9	81.2	67.5	79.4	66.1	76.8	7.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	94.5	71.5	64.1	92.2	70.8	62.5	87.1	69.5	33.5	82.0	32.3	79.6	31.2	76.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	18.2	16.3	DB	7.2	93.6	7.4	4.0	1.9	96.5	-2.4	98.8	-6.6	101.1	-11.9	104.0	(4)
(5)			WB	6.6	72.7	7.2	1.9	1.4	74.0	-2.8	75.1	-6.8	76.2	-12.1	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	32.4	33.5	40.6	50.5	56.9	63.7	67.7	66.1	59.4	50.7	42.2	36.4
	DBStd	14.31	8.92	8.39	8.30	7.59	7.33	6.69	6.31	6.05	6.18	7.08	6.90	7.81
	HDD50	2167	545	462	301	85	20	1	0	0	3	80	246	424
	HDD65	5763	1010	882	756	435	269	101	44	58	191	446	684	887
	CDD50	2202	1	0	10	101	233	412	549	499	284	100	12	1
	CDD65	324	0	0	0	2	17	62	129	92	21	1	0	0
	CDH74	2759	0	0	0	18	152	573	1129	753	131	3	0	0
	CDH80	830	0	0	0	1	25	156	383	246	19	0	0	0
Wind	WSAvg	8.1	9.7	8.9	8.9	7.8	7.5	6.9	6.8	6.5	7.1	8.1	9.0	9.6
Precipitation	PrecAvg	25.6	1.7	1.3	1.7	1.7	2.3	2.7	3.3	3.1	2.1	1.9	2.2	2.0
	PrecMax	34.9	4.8	2.8	4.7	4.9	6.0	8.0	7.4	9.2	5.9	6.3	4.7	6.5
	PrecMin	16.6	0.0	0.2	0.6	0.0	0.4	0.9	0.3	0.4	0.5	0.5	0.0	0.7
	PrecStd	4.6	0.9	0.7	1.0	1.0	1.3	1.4	1.8	2.0	1.1	1.2	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	53.9	66.2	76.8	82.8	89.5	93.2	92.8	82.7	73.2	60.9	55.0
		MCWB	49.3	46.1	51.7	57.5	63.7	68.0	66.2	66.9	64.3	59.7	55.2	48.9
	2%	DB	51.5	50.0	60.5	71.5	78.6	84.2	88.1	86.3	77.3	67.9	57.3	51.6
		MCWB	47.0	44.0	49.4	56.0	62.2	65.1	66.2	68.0	62.2	58.2	52.1	46.8
	5%	DB	47.8	46.8	55.7	68.0	73.7	80.6	84.2	81.0	73.7	64.3	54.0	48.6
		MCWB	44.6	42.0	47.5	54.8	59.4	64.5	65.2	65.4	61.5	56.4	49.4	45.1
	10%	DB	44.6	44.5	52.0	64.4	70.0	76.8	80.5	77.3	70.0	60.9	51.9	46.4
		MCWB	41.6	40.4	45.7	52.8	57.5	63.0	64.7	63.9	60.1	55.2	47.6	43.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	48.0	53.5	59.8	66.2	71.6	70.5	70.7	67.4	61.5	55.7	50.2
		MCDB	53.7	51.2	61.9	72.7	79.6	81.3	83.3	83.2	77.0	69.4	59.9	53.4
	2%	WB	46.8	44.9	50.8	57.4	63.7	68.1	68.4	68.6	65.0	59.5	52.6	47.5
		MCDB	50.5	48.5	58.1	68.6	75.0	80.5	81.2	80.8	73.9	65.9	56.9	50.9
	5%	WB	44.0	42.3	48.9	55.4	61.4	65.8	67.3	67.3	62.9	57.3	50.3	45.3
		MCDB	46.7	46.1	54.9	65.3	70.7	76.8	79.6	78.5	70.5	62.8	53.7	48.0
	10%	WB	41.4	40.2	46.3	53.7	59.5	63.8	66.0	65.6	61.0	55.4	48.3	43.3
		MCDB	43.7	43.7	51.3	63.4	68.1	73.7	77.0	74.8	68.2	60.5	51.4	45.8
Mean Daily Temperature Range	5% DB	MDBR	7.9	9.6	12.7	16.3	16.5	17.0	17.2	16.7	14.8	12.0	8.9	7.9
		MCDBR	11.0	13.6	19.8	22.3	22.3	23.3	23.7	23.0	20.1	16.9	11.7	10.5
		MCWBR	8.5	9.2	11.5	11.0	10.2	9.9	7.8	8.4	9.1	9.1	8.0	7.8
	5% WB	MCDBR	10.4	12.7	18.1	20.6	19.6	20.5	19.6	20.0	17.6	15.7	10.9	9.7
		MCWBR	8.5	9.1	11.1	10.9	9.8	9.7	7.9	8.7	9.1	9.4	8.1	7.7
Clear-Sky Solar Irradiance	taub	0.309	0.333	0.376	0.418	0.420	0.415	0.432	0.426	0.387	0.362	0.334	0.304	
	taud	2.362	2.338	2.250	2.183	2.218	2.260	2.221	2.255	2.338	2.385	2.402	2.381	
	Ebn at Noon	224	250	258	259	265	267	260	255	252	236	213	207	
	Edh at Noon	20	27	35	42	43	41	42	39	32	25	19	17	
All-Sky Solar Radiation	RadAvg	241	470	785	1273	1547	1656	1570	1367	981	571	272	184	
	RadStd	37	78	106	167	184	137	189	126	129	71	38	21	